

Static Timing Analysis

Sci.D.Professor Vazgen Melikyan

Synopsys Armenia Educational Department

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Course Overview

1. STA Concepts
 - 1 lecture
2. Delay Modeling
 - 1 lecture
3. Interconnect Parasitics
 - 1 lecture
4. Delay Calculation
 - 1 lecture
5. Configuring the STA Environment
 - 1 lecture
6. Timing Checks
 - 1 lecture
7. Crosstalk and Noise
 - 1 lecture

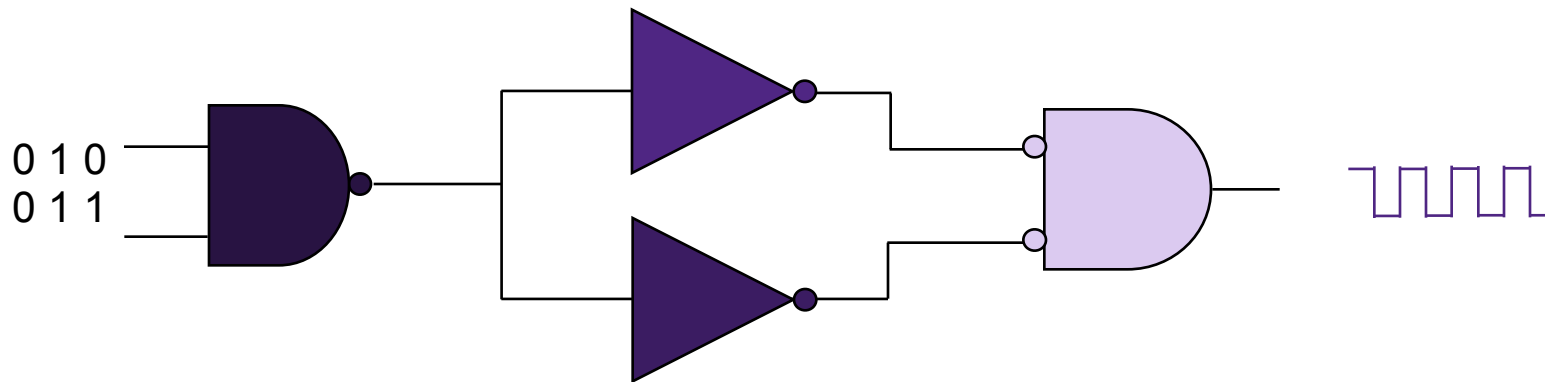
1. STA Concepts

Static Timing Analysis (STA)

- STA is a technique for digital design verification. It:
 - Validates if the design can operate at the set timing constraints
 - Is a complete and exhaustive verification of all timing checks of a design
 - Is used instead of simulation

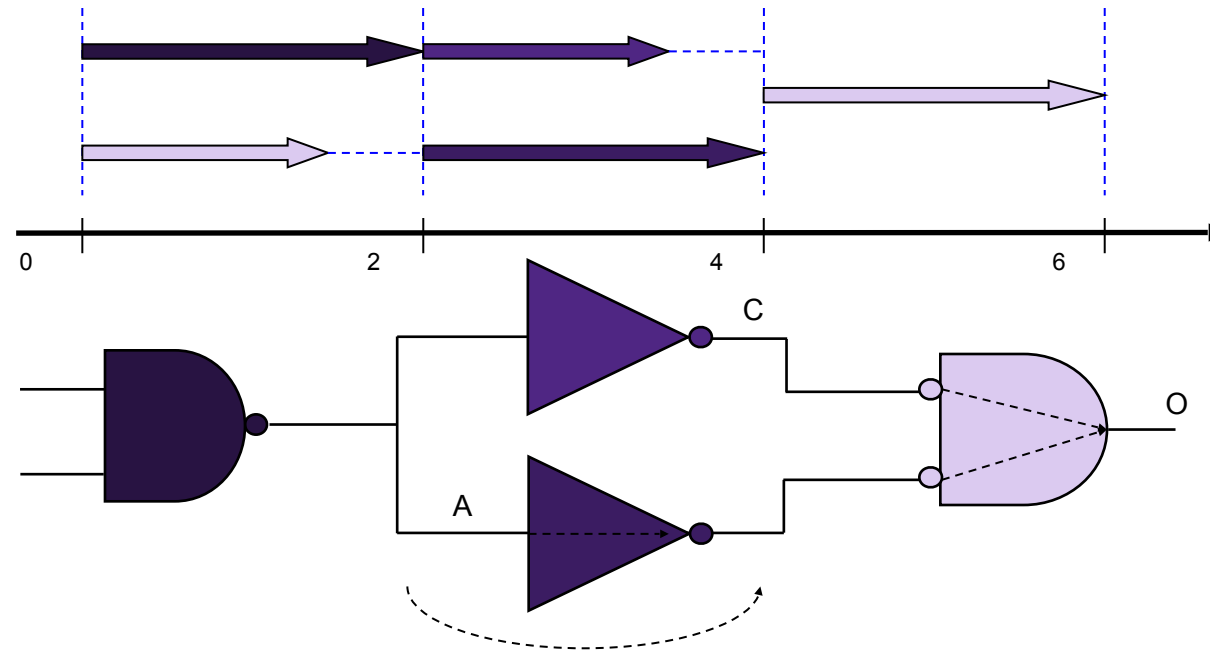
Simulation

- Simulating (dynamically) circuit response for a specified set of input patterns
 - Circuit modeled as network of capacitors, resistors and voltage/current sources



STA (2)

- Vector-less topological analysis of a circuit.
 - The signal at the input is propagated through the gates at each level till it reaches the output



STA vs. Simulation

- Simulation
- Advantages
 - Can be very accurate (Spice-level)
- Disadvantages
 - Analysis quality depends on stimulus vectors
 - Non-exhaustive, slow
- Static Timing Analysis (STA)
- Advantages
 - Exhaustive timing coverage
 - Does not require input vectors
 - Faster operation
- Disadvantages
 - Less accurate
 - Must define timing requirements/exceptions
 - Difficulty in handling asynchronous designs, false paths

Digital Circuits Timing Goals

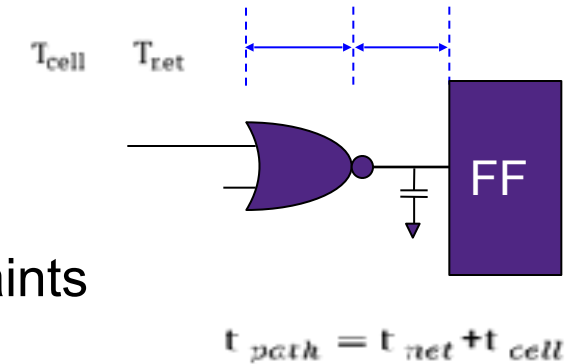
- Achievement of required frequency
 - Digital circuits are constrained to work under specific frequencies
 - All separate parts of circuit are constrained to have delay smaller than clock period
- Meeting timing constraints
 - Avoiding collision of signals
 - Avoiding failure
- STA verifies these

Problem Variability

- Other variables affect circuit timing, thus need to be considered during design
 - Operating Conditions
 - ♦ Process, Voltage, Temperature variations
 - Unstable clock frequency (jitter, skew)
 - ♦ Instability of clock frequency requires design margin
 - On-chip variation (OCV)
 - ♦ Device/Interconnect

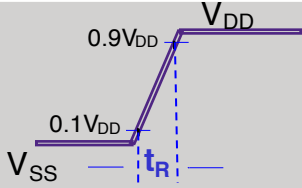
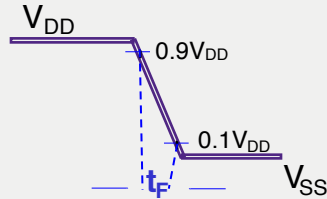
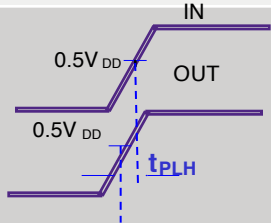
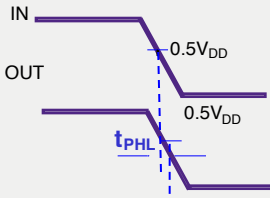
Components of Circuit Timing

- Delay components
 - Cells, Interconnects
- Constrained components
 - Clocked registers require setup/hold, recovery/removal constraints



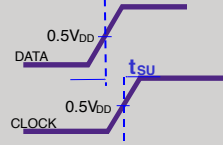
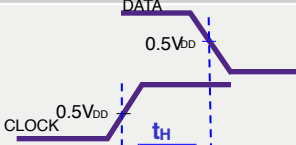
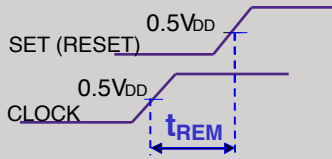
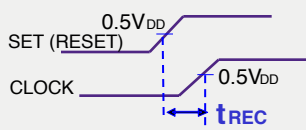
For circuit to operate without failure it is **required** that delay components at no point violate constraints of other components.

Cell Timing Parameters

N	Parameter	Unit	Symbol	Figure	Definition
1.	Rise transition time	ns	t_R		The time it takes a driving pin to make a transition from kV_{DD} to $(1-k)V_{DD}$ value. Usually $k=0.1$ (also possible $k=0.2, 0.3$, etc)
2.	Fall transition time	ns	t_F		The time it takes a driving pin to make a transition from $(1-k)V_{DD}$ to kV_{DD} value. Usually $k=0.1$ (also possible $k=0.2, 0.3$, etc)
3.	Propagation delay low-to-high (rise)	ns	t_{PLH} (t_{PR})		Time difference between the input signal crossing a $0.5V_{DD}$ and the output signal crossing its $0.5V_{DD}$ when the output signal is changing from low to high
4.	Propagation delay high-to-low (Fall)	ns	t_{PHL} (t_{PF})		Time difference between the input signal crossing a $0.5V_{DD}$ and the output signal crossing its $0.5V_{DD}$ when the output signal is changing from high to low

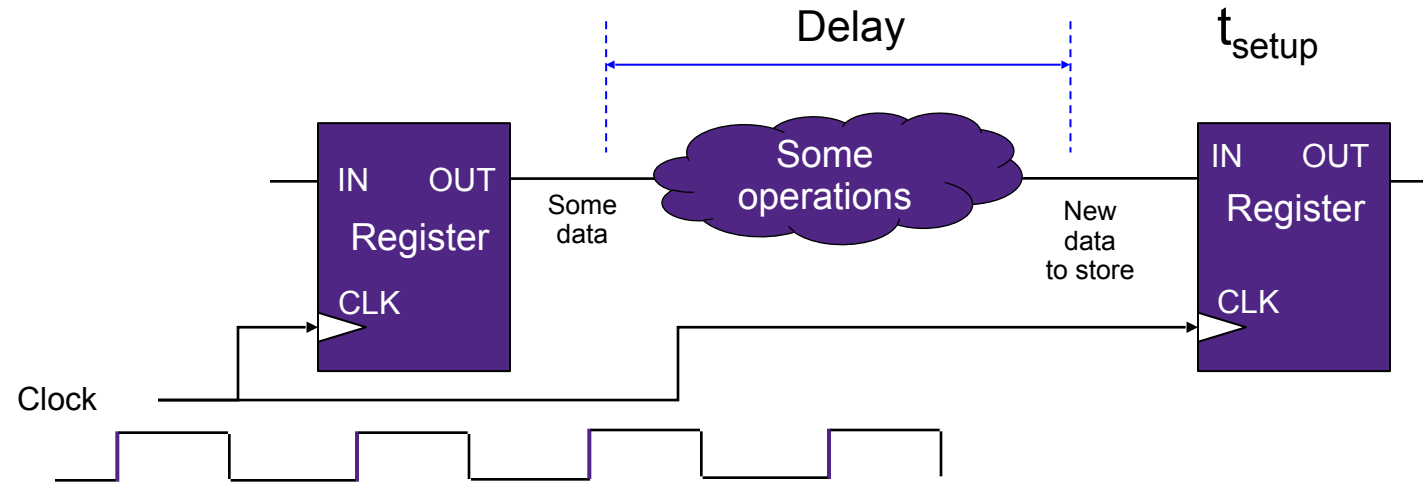
Cell Timing Constraints

- Setup/Hold, Recovery/Removal Constraints

N	Parameter	Unit	Symbol	Figure	Definition
1	Setup time (only for flip-flops or latches)	ns	t_{SU}		The minimum period in which the input data to a flip-flop or a latch must be stable before the active edge of the clock occurs
2..	Hold time (only for flip-flops or latches)	ns	t_H		The minimum period in which the input data to a flip-flop or a latch must remain stable after the active edge of the clock has occurred
3	Removal time (only for asynchronous Set or Reset)	ns	t_{REM}		The minimum time in which the asynchronous Set or Reset pin to a flip-flop or latch must remain enabled after the active edge of the clock has occurred
4	Recovery time (only for asynchronous Set or Reset)	ns	t_{REC}		The minimum time in which Set or Reset must be held stable after being deasserted before next active edge of the clock occurs

Timing Closure Problem

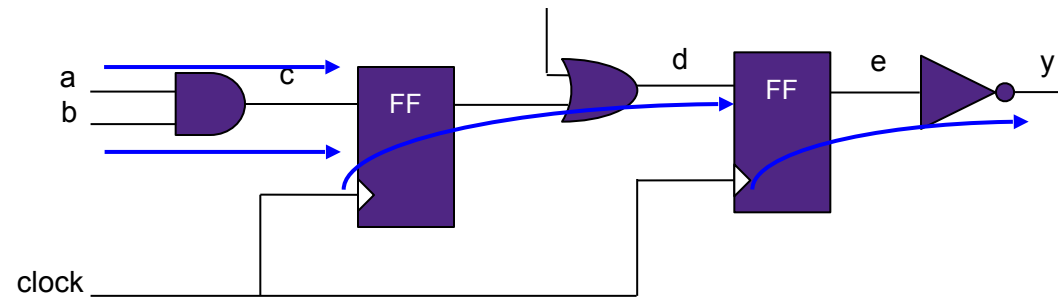
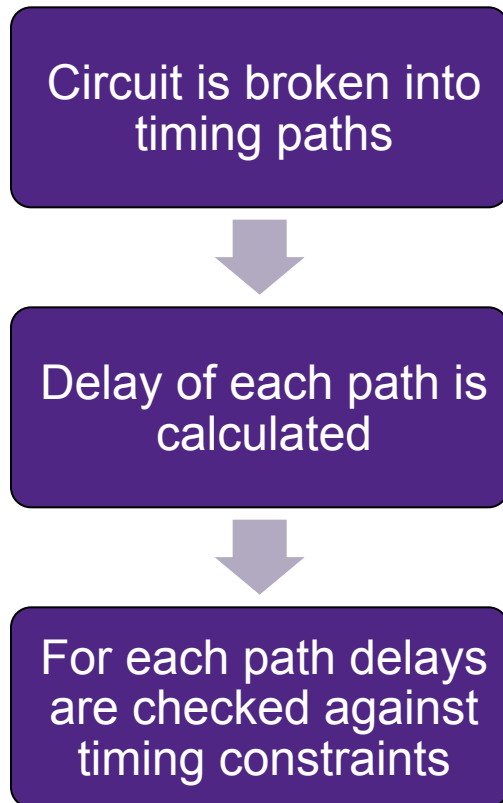
- Problem
 - In clocked environment signals on the register inputs must arrive before next reading sequence



Clock: Both registers store new value at specific points of time (with some frequency, like 1GHz)

$$\text{Delay} + t_{\text{setup}} < T_{\text{clock}}$$

Static Timing Analysis Steps



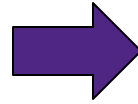
Path	Delay	Constraint
Path: a, b, c, d, e, y a: 10, b: 10, c: 10, d: 10, e: 10, y: 10	?	?
Path: a, b, c, d, e, y a: 10, b: 10, c: 10, d: 10, e: 10, y: 10	?	?
Path: a, b, c, d, e, y a: 10, b: 10, c: 10, d: 10, e: 10, y: 10	?	?
Path: a, b, c, d, e, y a: 10, b: 10, c: 10, d: 10, e: 10, y: 10	?	?

STA Concepts

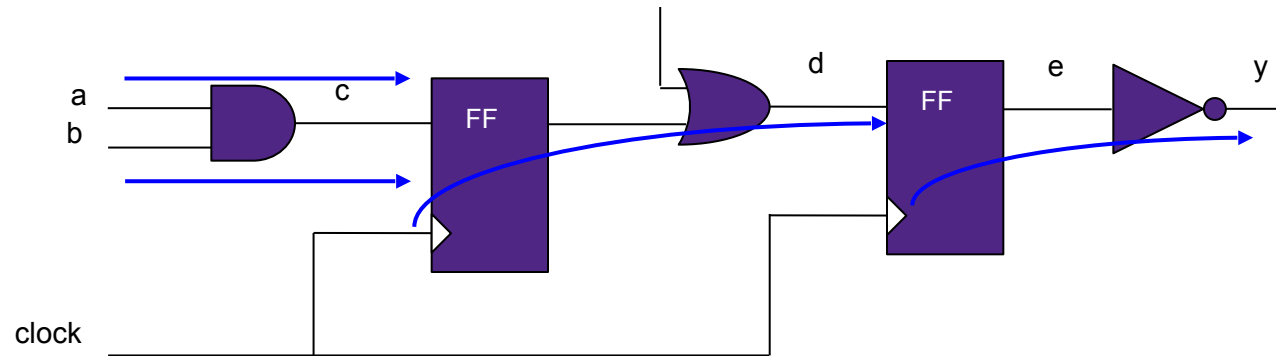
- Timing Path
- Required Time
- Arrival Time
- Slack
- Critical Path

Timing Paths

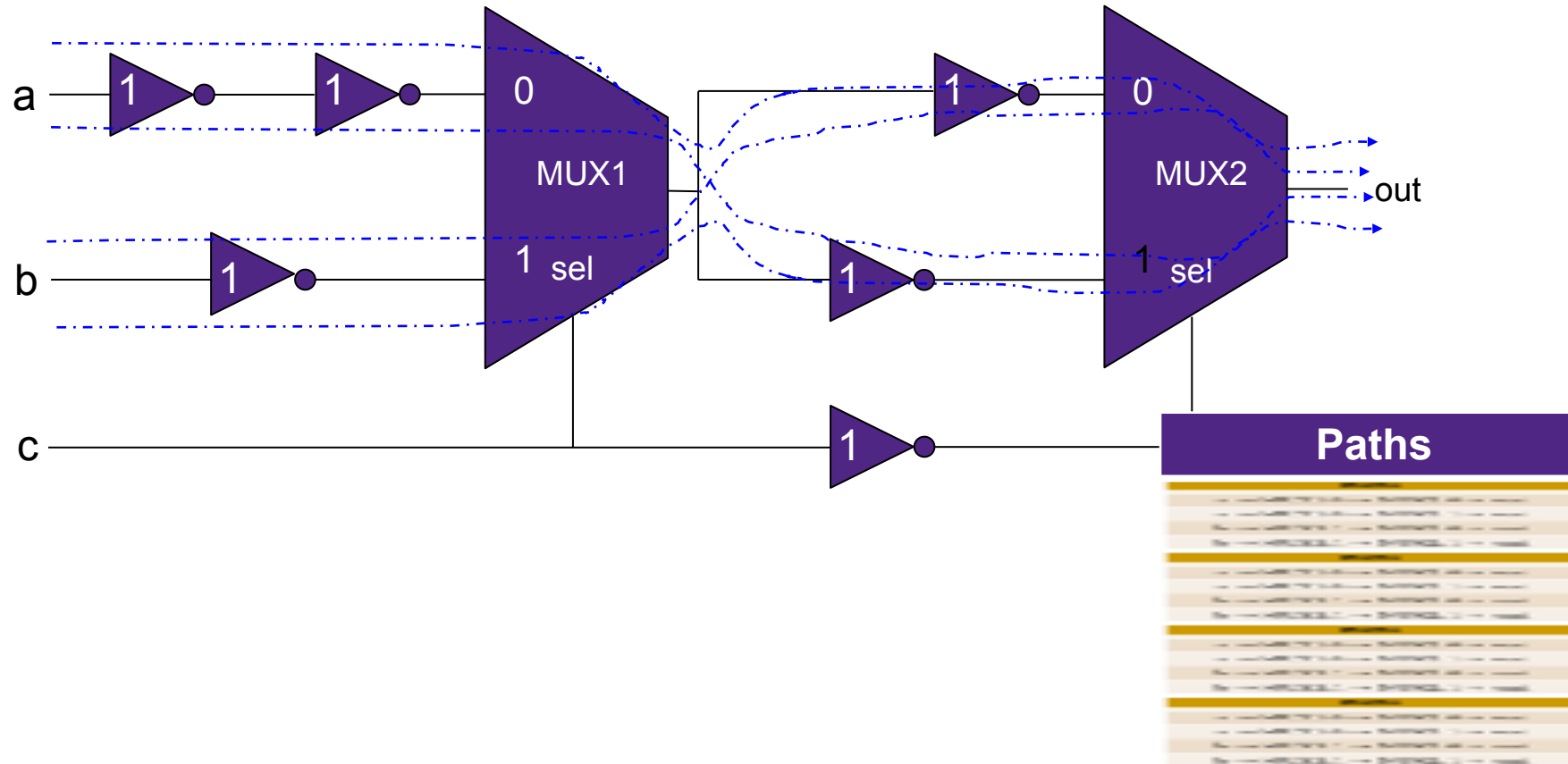
- Timing path has a start point and an endpoint.
- Start point:
 - Input ports
 - Clock pins of flip-flops



- Endpoints:
 - Output ports
 - data input pins of flip-flops

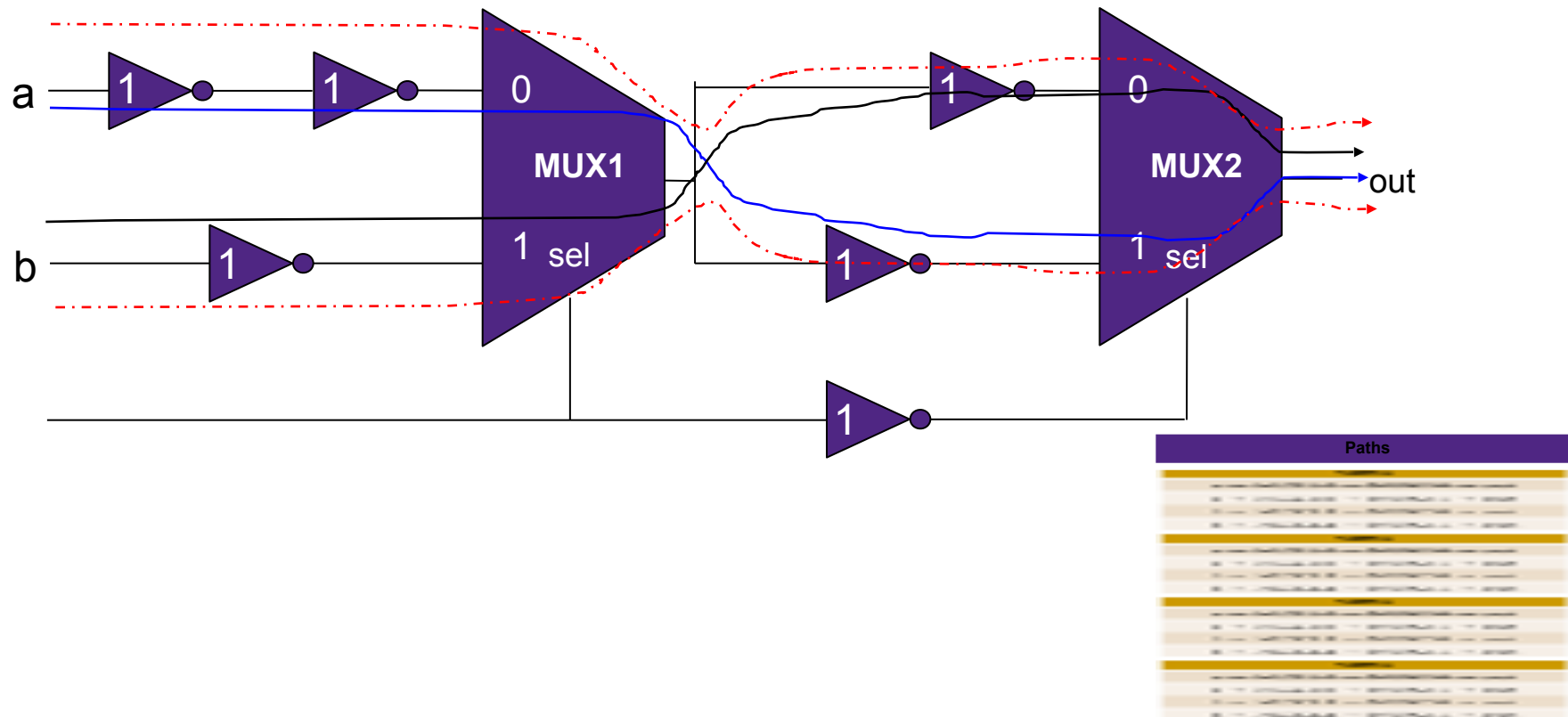


Possible Paths



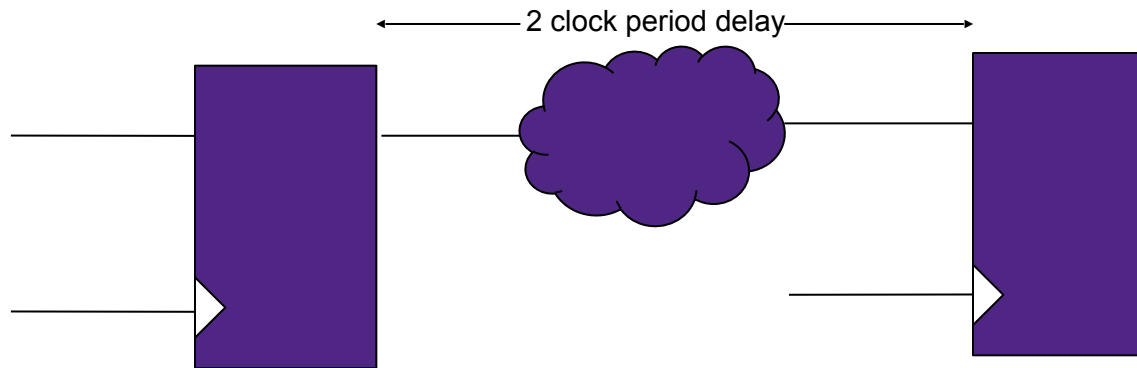
False Paths

- Paths which physically exist in a design but are not logic paths. These paths never get sensitized under any input condition



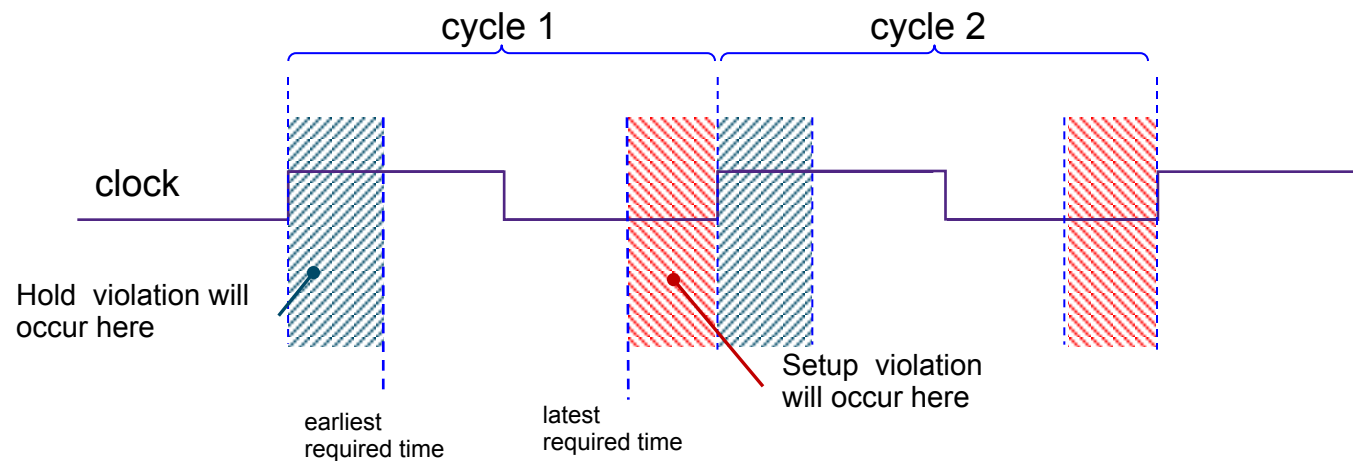
Multi-cycle Paths

- There are data paths that require more than one clock period for execution.



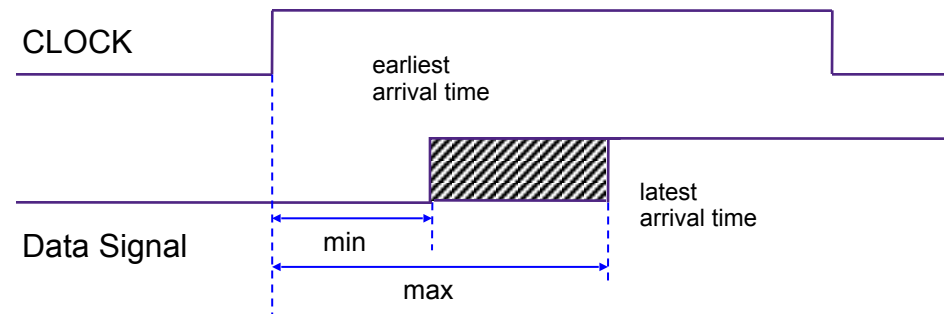
Required Time

- **Required time** specifies the time point (interval) at which data is required to arrive at end point (data is required to be stable after arrival).
 - Time point after which data can become unstable (change) is called earliest required time
 - Time point after which data cannot become unstable (change) is called latest required time
- The requirement is set by timing constraints like setup/hold, removal/recovery, etc.



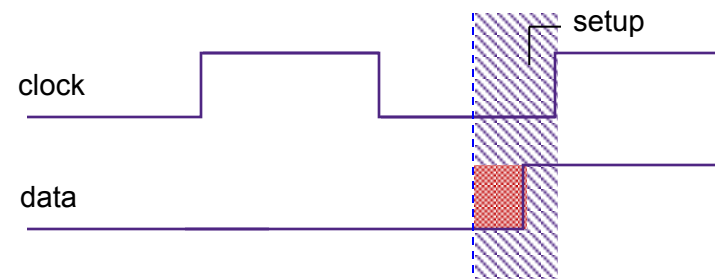
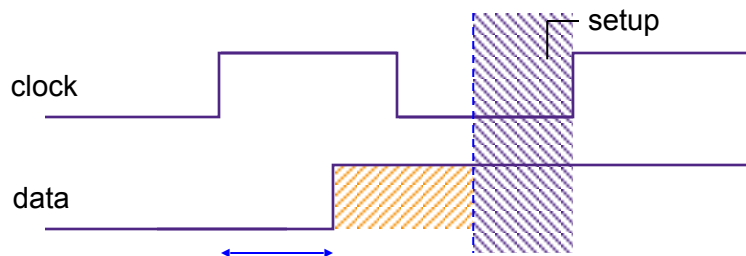
Arrival Time

- Arrival time defines the time interval during which a data signal will arrive at a path endpoint (after arrival time signal will be stable).
- Data is normally triggered by clock edge
- Data arrival depend on circuit delay, which vary (depend on temperature, supply voltage, etc.)
- Minimum delay, early arrival
- Maximum delay, late arrival



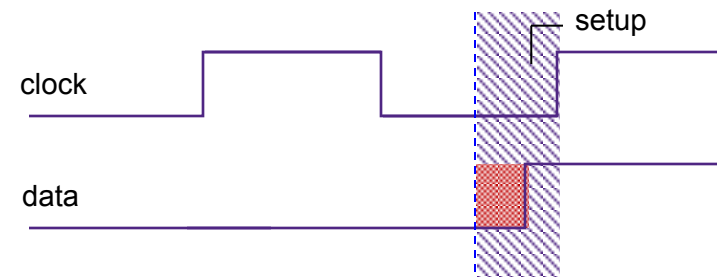
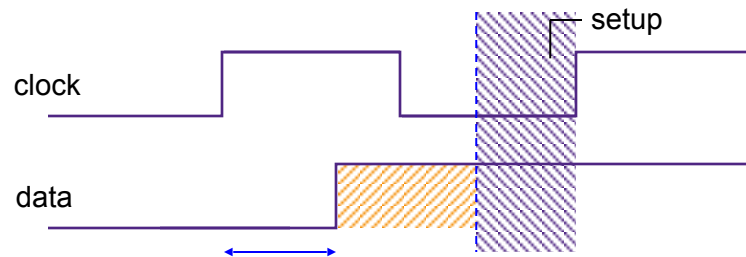
Slack and Critical Path

- **Slack** is the difference between the required time and the arrival time.
 - Negative slack → violation
 - Positive slack → constraints have been met
 - Critical path is a path in the design that has smallest slack



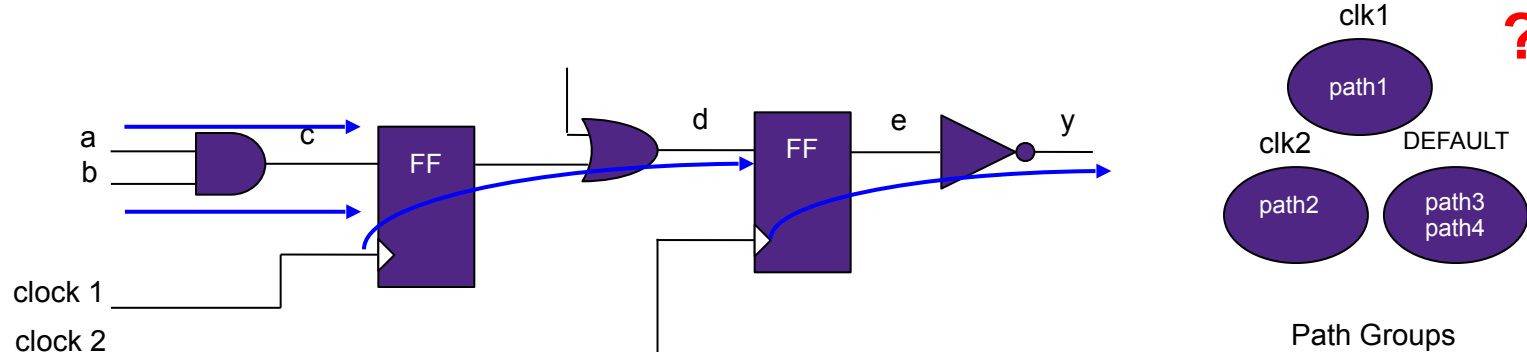
Early and Latest Analysis

- STA tool calculates the slack of each logic path, in order to find critical path.
- Early and Latest analysis approaches:
 - Assumes circuits have minimum delay, compares arrival time to earliest required time (hold check)
 - Assumes circuits have maximum delay, compares arrival time to latest required time (setup check)



Path Groups

- Timing paths are grouped into path groups by the clocks controlling their endpoints
- Slack is calculated in relation to each clock



Thank You